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GLOSSARY ON USE AND CULTIVATION OF COMMON MEDICINAL HERBS



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Aloe vera



Scientific name	:	*Aloe vera* (*Aloe barbedensis*)
Common name	:	Aloe (English), Grithkumari (Hindi), Lolesara (Kannada), Kumari (Sanskrit)
Growth Habit	:	Small herbaceous perennial plant with thick fleshy lanceolate leaves in the form of a rosette
Economic Part	:	Leaf, gel and juice
Medicinal Uses	:	Aloe gel has the property of relieving thermal burn and sunburn as well as promoting wound healing. It also has moisturizing and emollient properties. Aloin is used in anti-obesity preparation. Gel is also used in flavouring beverages, fruit juices and smoothies
Herbal Preparations	:	Aloe gel, juice, RTS

Cultivation practices

Varieties	:	CIM-Sheetal
Climate	:	Warm humid or dry climate with 150-200 cm to about 35-40 cm yearly rainfall during the growing period
Soil	:	Sandy coastal soils to loamy soils of plains with pH up to 8.5
Propagation	:	Root suckers / rhizome cuttings (28,000 - 34,000 suckers / ha)
Spacing and planting	:	15-18 cm long root suckers or rhizome cuttings are planted at 60 x 30 cm or 60 x 45 cm. Planting time is during March to June
Manuring	:	25 tonnes of FYM/ha
Harvesting and Yield	:	Commercial yield starts from 2 years to 5 years of age. Long mature leaves from the plant base are to be harvested for high gel yield. Average fresh leaf yield is around 10-12 t/ha.
Selling price	:	The average market price of Aloe leaf ranges from Rs 60-100/kg

Ashwagandha



Scientific name	:	*Withania somnifera*
Common name	:	Indian ginseng (English), Asgandhi (Hindi), Kiremaddinagida (Kannada), Kamrupini (Sanskrit)
Growth Habit	:	Erect herbaceous evergreen annual to perennial, branched, under shrub of about 30 - 120 cm height
Economic Part	:	Roots
Medicinal uses	:	Reduce anxiety and stress, help fight depression, boost fertility in men, boost brain function. Used as anti- inflammatory, diuretic, narcotic, abortifacient
Herbal Preparations	:	Ashwagandha churna, Ashwagandha tea, Ashwagandha tablets

Cultivation practices

Varieties	:	Arka Ashwagandha, Jawahar Asgandh-20, Jawahar Asgandh-134, Poshitha
Climate	:	Prefers dry weather for its successful growth, grown as rainfed crop
Soil	:	Sandy loam or light red soils with good drainage and soil pH range of 7.5 to 8 is ideal
Propagation	:	Seeds (10-12 kg/ha)
Spacing and planting	:	Direct sowing by broadcasting during late rainy season (September) is preferred for better plant stand
Harvesting and Yield	:	Crop is ready for harvest 150-180 days after sowing. An average yield of 300 to 500 kg of dried roots and 50 to 75 kg seeds can be obtained from one hectare area.
Selling price	:	Average market price of dried roots is Rs. 300/kg

Bhringaraj



Scientific name	:	*Eclipta alba*
Common name	:	False Daisy (English), Bhringaraj (Hindi), Garugada soppu (Kannada), Bhringaraj (Sanskrit)
Growth Habit	:	Annual, erect or prostrate, branched herb
Economic Part	:	Whole herb
Medicinal Uses	:	Herb is used as a tonic and deobstruent in enlargement of the liver and spleen and in various chronic skin diseases. Fresh juice of the leaves is used as hair restorer by rubbing it on the shaven scalp and also improves the colour of the hair
Herbal Preparations	:	Bhringraj churna, Bhringaraj oil, Bhringraj powder

Cultivation practices

Varieties	:	No named varieties
Climate	:	Comes up well in tropical, sub-tropical and temperate regions. It prefers warm climate with a temperature range of 25 °C to 35 °C.
Soil	:	Prefers red loamy soils with high organic matter and high moisture content
Propagation	:	Seeds (400-500 g/ha) or stem cuttings (25000 nos/ha).
Spacing and planting	:	Seeds raised in nursery are transplanted to the main field at spacing of 20 x20 cm when they are 45-60 days old. Alternately, 10-15 cm long terminal cuttings with 5-6 nodes can be used.The best time for nursery sowing is February–March or rainy season
Manuring	:	20 tonnes of FYM/ha
Harvesting and Yield	:	Harvesting is done 90 days after transplanting by cutting the above ground portion leaving behind 10-12 cm for ratoon crop. It gives about 8 t/ha fresh herbage yield during 1st cropping (April-July) and 3 t/ha from the ratoon crop (August- September)
Selling price	:	The average market price for dry herb is Rs 25 /kg

Bhuiamlaki



Scientific name	:	*Phyllanthus amarus*
Common name	:	Country gooseberry (English), Jaramala (Hindi), Nelanelli (Kannada), Bhudhatri (Sanskrit)
Growth Habit	:	Erect glabrous annual herb branching at the base
Economic Part	:	Whole herb
Medicinal Uses	:	Useful in Hepatitis B and Jaundice, intestinal infection, diabetes and essential ingredient of many indigenous poly herbal formulations used in the recovery of bronchitis, leprosy, anaemia, asthma and hiccough
Herbal Preparations	:	Bhumi amla juice, Bhumi amla churna

Cultivation practices

Varieties	:	Navyakrit, CIM-Jeevan
Climate	:	Prefers tropical high rainfall conditions
Soil	:	Prefers clayey to loamy soils with soil pH ranging from 5.5 to 8
Propagation	:	Seeds (1 kg/ha)
Spacing and planting	:	Seeds are raised in nursery and 35-40 days old seedlings of 10-15 cm height are transplanted in the field at 15 x 10 cm spacing during June-July.
Manuring	:	10 tonnes of FYM/ha
Harvesting and Yield	:	The crop is ready for harvest after 3 months of transplanting. An average yield of 2000 kg of dry herb can be obtained per hectare
Selling price	:	The average market price for dry herb is Rs 25 /kg

Brahmi



Scientific name	:	*Bacopa monnieri*
Common name	:	Water hyssop (English), Brahmi (Hindi), Nirubrahmi (Kannada), Saraswati (Sanskrit)
Growth Habit	:	Perennial, succulent, glabrous, creeping herb, with rooting at nodes
Economic Part	:	Whole herb
Medicinal Uses	:	Intellect promoter, hypotensive, and neuropathic sedative drug. It is a potent tonic used to improve mental alertness, learning performance, and memory as well as in the treatment of insomnia.
Herbal Preparations	:	Brahmi fresh juice, Brahmi churna, Brahmi paste, Brahmi infusion

Cultivation practices

Varieties	:	Subodhak, Pragyashakti, and CIM-jagriti
Climate	:	Crop prefers warm humid tropical to sub tropical climate. A temperature range between 33-40 °C with relative humidity of 60-65% has been found good for optimum vegetative growth
Soil	:	Clayey loam to clayey soils with soil pH (5 - 7.5) is best suited for the growth
Propagation	:	Soft wood stem cuttings or runners (70 kg material i.e 40000 propagules/ha)
Spacing and planting	:	5-6 cm long cuttings each with a few leaves and nodes are transplanted in wet soil at spacing of 10 x 10 cm. The cuttings should be transplanted in the month of July-August to obtain maximum herbage yield.
Manuring	:	10 tonnes of FYM/ha
Harvesting and Yield	:	Herb is harvested after 6 months of planting. On an average, a yield of 300 q/ ha fresh and 60 q/ha of dry herbage can be obtained from a single harvest. After the first harvest, 40 q/ha additional dry herbage yield can be achieved from the ratoon crop.
Selling price	:	Average market price of whole herb is Rs. 216/kg.

Garden mint



Scientific name	:	*Mentha arvensis*
Common name	:	Japanese mint (English), Pudina (Hindi, Kannada, Sanskrit)
Growth Habit	:	Perennial creeping herb rooting at the nodes
Economic Part	:	Whole herb
Medicinal Uses	:	Used for flavouring in food and pharmaceutical industries. Helps in digestion, used in the treatment of nasal and bronchial catarrh; it is also occasionally employed as a carminative, and as an analgesic in the treatment of neuralgia and rheumatism
Herbal Preparations	:	Mint soup, Mint tea, Mint powder

Cultivation practices

Varieties	:	MAS-1, Kalka/ Hyb 77, Shivalik, Gomti, Himalaya, Kosi, Damroo, Saksham, Sambhav, Kushal, CIMAP-Patra
Climate	:	Tropical and sub-tropical areas with temperature in the range 25-30 °C
Soil	:	Medium to fertile deep soil, rich in humus with good water-holding capacity and pH in the range of 6-7.5 is ideal.
Propagation	:	Stolon or suckers (400 kg/ha)
Spacing and planting	:	7-10 cm long stolon pieces are planted during December-January in shallow furrows at 45-60 cm row to row spacing
Manuring	:	25-30 tonnes of FYM/ha
Harvesting and Yield	:	First harvest is after 100-120 days of planting when the lower leaves start turning yellow and subsequent harvest is done about 80 days after the first harvest and the third one after about 80 days from the second harvest. The average yield of mint from three cuttings is 20-25 tonnes per hectare.
Selling price	:	Average market price of dried leaves is Rs 100/kg

Giloe



Scientific name	:	*Tinospora cordifolia*
Common name	:	Heart leaved moon seed (English), Giloe (Hindi), Amruthaballi (Kannada), Amrita (Sanskrit)
Growth Habit	:	Perennial woody climber with succulent stems
Economic Part	:	Stem and leaves
Medicinal Uses	:	Anti-diabetic, anti-periodic, anti-spasmodic, anti-inflammatory, anti-arthritic, anti-oxidant, anti-allergic, anti-stress, anti-leprotic, anti-malarial, hepatoprotective, immunemodulatory and anti-neoplastic activities
Herbal Preparations	:	Giloe juice, Giloe churna, Giloe kwath, Giloe tablet

Cultivation practices

Varieties	:	No named varieties
Climate	:	Prefers warm and humid tropical climate, with an annual rainfall of 2000-3000 mm.
Soil	:	Prefers medium black or red acidic to neutral soils
Propagation	:	Rooted stem cuttings (10,000 cuttings/ha)
Spacing and planting	:	10-15 cm long stem cuttings with two nodes are planted in nursery beds and transplanted to the main field after 10 weeks at a spacing of 1x1m. Planting time is during the onset of monsoon in May-June.
Manuring	:	10 tonnes of FYM/ha
Harvesting and Yield	:	First harvest of leaves is obtained after 3-4 months of planting and subsequently at two months interval. An average yield of about 5000 kg of dried leaves can be obtained from one hectare in a year.
Selling price	:	Average market price for dried stem is Rs 38/kg

Gudmar



Scientific name	:	*Gymnema sylvestre*
Common name	:	Australian cowplant (English), Gurmar (Hindi), Madhunashini (Kannada & Sanskrit)
Growth Habit	:	Perennial, laticiferous, pubescent, evergreen woody climber
Economic Part	:	Leaves
Medicinal Uses	:	Anti-diabetic, anti-obesity, astringent, stomachic and tonic. The leaf powder stimulates heart and circulatory system and increases secretion of urine.
Herbal Preparations	:	Gudmar leaf powder, Gudmar kwath, Gudmar churna

Cultivation practices

Varieties	:	No named varieties
Climate	:	Prefers tropical, sub-tropical type of climate and also dry areas. Areas with high or medium well distributed rainfall are suitable.
Soil	:	Red sandy loam or medium deep black soil is ideal for this crop
Propagation	:	Seeds (2-3 kg/ha) or rooted stem cuttings
Spacing and planting	:	Seeds are raised in nursery and transplanted to main field after 40-50 days. 15 cm long matured stem cuttings raised in polybags are transplanted in the pits after 90 days at a spacing of 2.5 x 1.8 m. Best season for planting is during the month of June-July
Manuring	:	10 kg FYM per pit
Harvesting and Yield	:	Harvesting of leaves is done 1 year after planting. On an average 5-6 kg dried leaves per plant can be obtained from a 3-4 years old vine and accounting about 11-12 tonnes of dried leaves per hectare.
Selling price	:	Average market price for dried leaf is Rs 128/kg

Hadjod



Scientific name	:	*Cissus quadrangularis*
Common name	:	Veldt grape, Adamant creeper, Devil's backbone (English), Asthisonhara / Vajravalli (Sanskrit), Hadjod / Hadjora (Hindi), Mangaravalli (Kannada)
Growth Habit	:	It is a quadrangular, succulent shrubby climber with 4 winged internodes and a tendril at the nodes. Stem joined at nodes, internodes are 8 to 10 cm long and 1.2 to 1.5 cm wide. Flowers are small, greenish white and opposite to the leaves.
Economic Part	:	Whole plant (leaf, stem and root)
Medicinal Uses	:	Used for bone health, repair bone fractures, relieve joint pain, regulate menstruation and protect against chronic conditions like heart disease, diabetes, stroke, hemorrhoids, gout, asthma, weight loss and allergies
Herbal Preparations	:	Asthisamharaka juice, powder and decoction

Cultivation practices

Varieties	:	No named varieties
Climate	:	It prefers warm tropical climate, up to an elevation of 500 m
Soil	:	The plant grows in almost all type of soil with better water drainage system. It prefers loamy soil with pH range of 5.5 to 7.5.
Propagation	:	Mature stem cuttings of 30 cm long and having two nodes are commercially used
Spacing and planting	:	Stem cuttings are planted in a pit of 15×15×15 cm at a spacing of 30 × 30 cm during kharif season (June-July). The pits are filled with a mixture containing vermi-compost: FYM : sand : bone meal in equal proportions. The plants are to be supported with the help of bamboo sticks.
Manuring	:	Farm yard manure is applied (10-12 tonnes/ha) after planting
Harvesting and Yield	:	It is a perennial crop. Stems are cut and air dried 11 months after planting. Whole plant can be harvested after 2 years of planting.
Selling price	:	NA

Indian Borage



Scientific name	:	*Coleus aromaticus/ amboinicus; Syn: Plectranthus amboinicus*
Common name	:	Country borage, Indian borage (English), Patta ajavayin, Patharchur (Hindi), Dodda pathre, Karpurahalli, Doddapathre (Kannada), Karpuravalli, Sugandhavalakam (Sanskrit)
Growth Habit	:	Large, succulent, highly aromatic, pubescent perennial herb, 30-90 cm in height and with thick fleshy stem and leaves.
Economic Part	:	Leaves
Medicinal Uses	:	Used to treat malarial fever, hepatopathy, renal and vesical calculi, cough, chronic asthma, hiccough, bronchitis, helminthiasis, colic, convulsions, and epilepsy
Herbal Preparations	:	Food flavouring, dhal, chutney

Cultivation practices

Varieties	:	No named varieties
Climate	:	Tropical and sub-tropical areas with temperature in the range 28-35 °C and mean annual rainfall in the range 2,000 - 2,600 mm, elevations up to 1,500
Soil	:	Medium to fertile well drained soil, rich in humus with good water-holding capacity and pH in the range of 6-6.5 is ideal.
Propagation	:	Commercially propagated through stem cuttings.
Spacing and planting	:	Rooted stem cuttings has to be planted at the onset of monsoon at 45-60 cm row to row spacing
Manuring	:	20 tonnes of FYM/ha
Harvesting and Yield	:	The leaves are harvested as required and used fresh.
Selling price	:	Not available

Insulin Plant



Scientific name	:	*Costus igneus*
Common name	:	Spiral flag, Blazing *Costus*, Step stepping stool (English), Insulin ele (Kannada)
Growth Habit	:	It is a perennial, upright, spreading plant reaching about two feet tall, with spirally arranged leaves which are alternate, oblong, twenty to twenty-five cm in length with several parallel thick veins and orange colour attractive flowers. Fruits are very small, green in color.
Economic Part	:	Leaves
Medicinal Uses	:	Anti-diabetic, hypolipidemic, diuretic, antioxidant, anti-microbial, anti-cancerous
Herbal Preparations	:	Leaf powder

Cultivation practices

Varieties	:	No named varieties
Climate	:	It grows under full sun or partial shade.
Soil	:	It needs fertile soil and ample moisture, and is often planted near water
Propagation	:	Propagation occurs by the division of the clumps, cuttings, or by separating the offsets that form below the flower heads.
Spacing and planting	:	Rooted stem cuttings has to be planted at the onset of monsoon at 45-60 cm row to row spacing
Manuring	:	20 tonnes of FYM/ha
Harvesting and Yield	:	Mature leaves are harvested as and when required and used fresh or powdered form
Selling price	:	Not available

Kalmegh



Scientific name	:	*Andrographis paniculata*
Common name	:	King of bitters (English), Kalmegh (Hindi), Nelabevu (Kannada), Yavatikta (Sanskrit)
Growth Habit	:	Annual herbaceous or shrubby procumbent or erect plant
Economic Part	:	Whole plant
Medicinal Uses	:	Used in liver problems, to boost immunity and to manage the symptoms of the common cold, sinusitis and allergies due to its antimicrobial and immunomodulatory properties.
Herbal Preparations	:	Kalmegh churna, Kalmegh kwath, Kalmegh juice

Cultivation practices

Varieties	:	CIM Megha, Anand Kalmegh-1
Climate	:	Prefers moist, shady cooler climates with well distributed rainfall, also comes in dry forest areas.
Soil	:	Sandy loam soil rich in organic matter is good for its growth and yield
Propagation	:	Seeds (400 g/ha) or cuttings
Spacing and planting	:	Seeds are sown in nursery and transplanted to main field after 45-50 days of sowing at a spacing of 15x15 cm. Best season for planting this crop is during the month of June.
Manuring	:	25 tonnes of FYM/ha
Harvesting and Yield	:	Whole plant is harvested at 90-120 days after sowing. The average yield per hectare is about 2 to 2.5 tonnes of dry herb.
Selling price	:	Average market price of whole dry herb is Rs 47/kg.

Long pepper



Scientific name	:	*Piper longum*
Common name	:	Indian long pepper (English), Pipala (Hindi), Hippali (Kannada) Pippali (Sanskrit)
Growth Habit	:	Perennial slender creeping under shrub with betel like leaves
Economic Par	:	Spikes and roots
Medicinal Uses	:	Remedy for all sorts of Kapha aggravating disorders like asthma, bronchitis, cough and cold symptoms. It provides relief from cough and congestion and also helps in removing phlegm deposits from the respiratory tract. It also has carminative (reduces flatulence/gas), stomachic (promotes digestion) and laxative properties.
Herbal Preparations	:	Pippali choorna, Pippali decoction

Cultivation practices

Varieties	:	Viswam
Climate	:	Prefers heavy rainfall areas with high relative humidity. Can be successfully grown in hot moist climate with partial shade
Soil	:	Loamy and lateritic soils rich in organic matter and good water holding capacity is suitable
Propagation	:	Rooted cuttings and suckers
Spacing and planting	:	The rooted cuttings or suckers are planted in the pits at the rate of 2 per pit with the onset of monsoon at a spacing of 60 x 60 cm.
Manuring	:	20 tonnes of FYM/ha
Harvesting and Yield	:	Spikes are harvested 6 months after planting when the spikes turn blackish in colour. The average yield of roots is 500 kg/ha.
Selling price	:	The average market price of long pepper fruit is Rs 512/kg

Mandukaparni



Scientific name	:	*Centella asiatica*
Common nam	:	Asiatic pennywort (English), Brahma manduki (Hindi), Ondelaga (Kannada), Mandukaparni (Sanskrit)
Growth Habit	:	Perennial, glabrous small trailing herb rooting at the nodes
Economic Part	:	Whole herb
Medicinal Use	:	It is used as nervine tonic, for improving memory and mental disorders. It is used in anti-leprosy, diuretic, stomachic and used in insomnia, asthma, abdominal disorders and fever
Herbal Preparations	:	Centella fresh juice, Centella churna, chutney, green salad

Cultivation practices

Varieties	:	Arka Divya, Arka Prabhavi, Kayakriti, Majjaposhak, Vallabh medha
Climate	:	Tropical and subtropical mild climate with partial shade is ideal for its growth
Soil	:	Prefers acidic soils. Clayey soils with good moisture holding capacity along with organic matter suits very well for this crop.
Propagation	:	Rooted suckers (200 kg i.e 1,10,000 propagules/ha)
Spacing and planting	:	7-10 cm long cuttings with 3-4 nodes is planted at 30 x 30 cm spacing in the main field preferably with the onset of monsoon
Manuring	:	20 tonnes of FYM/ha (5t/ha FYM added after each harvest)
Harvesting and Yield	:	First harvest at 5 months after planting and subsequent harvest at 3 months interval. On an average, from one hectare area, about 5500 kg of fresh herb, 2000 kg of dry herb and 20 kg of asiaticoside can be obtained.
Selling price	:	Average market price of whole herb is Rs 97/kg

Makoi



Scientific name	:	*Solanum nigrum*
Common name	:	Black nightshade (English), Makoi, Makoy (Hindi), Kakamanchi (Kannada & Sanskrit)
Growth Habit	:	Annual herbaceous plant growing to 35-50 cm height
Economic Part	:	Whole plant
Medicinal Uses	:	Makoi can be used for the treatment of many ailments such as inflammation, ring worms, and can provide relief from burns, itching and pain. The herb can also be used for stomach irritation, cramps, spasms, pain, and nervousness.
Herbal Preparations	:	Fruit powder, fresh and dry fruits

Cultivation practices

Varieties	:	CO-1
Climate	:	Comes in tropical and subtropical areas up to an elevation of 2100 m
Soil	:	Well drained sandy loam soil is suitable
Propagation	:	Seeds (200-250g/ha)
Spacing and planting	:	Seeds are sown in nursery and seedlings are transplanted into the main field when they are 30 days old and are 10-15 cm in height having 6 leaves at spacing of 60 x 45 cm in June-July
Manuring	:	10-15 tonnes FYM/ha
Harvesting and Yield	:	The crop is ready for harvest after 3-4 months and it is harvested at mature green berry stage. The average dry herbage yield from this crop ranges from 6 to 8 t/ha
Selling price	:	The average market price of seeds is Rs 140/kg

Medicinal Coleus



Scientific name	:	*Coleus forskholii*
Common name	:	Medicinal coleus (English), Patharchur (Hindi), Makandiberu (Kannada), Pashan Bhedi (Sanskrit)
Growth Habit	:	Aromatic perennial plant 0.5 m tall with thick tubers
Economic Part	:	Tuberous roots
Medicinal Uses	:	Used against hypertension, glaucoma, asthma, cardiovascular diseases, obesity, congestive heart failures and cancer
Herbal Preparations	:	Forskholin extract, Forskholin capsules

Cultivation practices

Varieties	:	Aisiri, K-8
Climate	:	Prefers tropical humid and less humid regions with temperature ranging from 10-35 °C
Soil	:	Prefers red sandy loam porous and well-drained soils with pH ranging from 5.5 to 7
Propagation	:	Seeds, stem cuttings
Spacing and planting	:	Seeds sown in nursery are transplanted after 40 days (8-10 cm height) to main field. Alternately, 10-12 cm long cuttings comprising 3-4 pairs of leaves raised in nursery are transplanted after one month to main field. Seedlings or rooted cuttings are planted at 60 x 45 cm spacing in the month of June-July.
Manuring	:	20-25 tonnes FYM/ha
Harvesting and Yield	:	The crop is ready for harvest 5 to 6 months after planting. On an average, a yield of 1500-2000 kg of dry tubers may be obtained per hectare
Selling price	:	The average market price of dried roots is Rs 50/kg

Periwinkle



Scientific name	:	*Catharanthus roseus*
Common name	:	Periwinkle (English), Sadabahar (Hindi), Nityapushpa (Kannada), Sadapushpa (Sanskrit)
Growth Habit	:	Perennial evergreen shrub growing to 90-100 cm height
Economic Part	:	Leaves, stem, seeds and roots
Medicinal Uses	:	Hypotensive and antispasmodic properties, used to treat acute leukaemia, Hodgkin's disease and other lymphomas and chorio carcinomas
Herbal Preparations	:	Oncovin, Velbe, Vinca leukoblastine (commercial drugs)

Cultivation practices

Varieties	:	Nirmal, Dhawal, Prabal, Prabhat
Climate	:	Comes in tropical and sub-tropical areas up to an elevation of 1300 m from sea level. Well distributed rainfall of 100 cm or more is ideal
Soil	:	Deep sandy loam to loam soils of medium fertility is preferred
Propagation	:	Seeds (2.5 kg/ha for direct sowing; 500 g/ha for transplanting), cuttings
Spacing and planting	:	Seeds can be either directly sown or transplanted. Direct sowing should coincide with monsoon in the month of July. Transplanting is done for 2-month old seedling at a spacing of 45x30 cm. About 10-15 cm long soft-wood cuttings with minimum 5-6 nodes are ideal for planting.
Manuring	:	10-15 tonnes of FYM/ha
Harvesting and Yield	:	For leaves, plant is harvested 6 and 9 months after planting and for roots, plant is harvested 12 months after planting
Selling price	:	The average market price for dried leaves and root is Rs 100/kg

Sarpagandha



Scientific name	:	*Rauwolfia serpentina*
Common name	:	Indian snakeroot (English), Chhotachand (Hindi), Sarpagandha (Sanskrit)
Growth Habit	:	Perennial under shrub growing to a height of 60-90 cm
Economic Part	:	Roots
Medicinal Uses	:	Against high blood pressure and mental illnesses such as schizophrenia and anxiety.
Herbal Preparations	:	Sarpagandha churna, Sarpagandha kwath, Sarpagandha tablets

Cultivation practices

Varieties	:	RS-1, CIM-Sheel
Climate	:	Hot humid condition with temperature range of 10-30 °C in the open and partial shade conditions.
Soil	:	Sandy alluvial loam to red lateritic loam with soil pH 4.6-6.2
Propagation	:	Seeds (5.5 kg/ha), rooted cuttings (100 kg/ha)
Spacing and planting	:	Seeds are sown in nursery and 45-50 days old seedlings are transplanted in furrows at spacing of 45 x 30 cm. It can also propagated by root cuttings of 2.5-5 cm length or stem cuttings of 15-22 cm length having 3 internodes. Appropriate time of planting is from May-July
Manuring	:	25-30 tonnes FYM/ha
Harvesting and Yield	:	Roots are generally harvested from 18 months after planting. Average yield of 2,200 kg of air-dried roots/ha is obtained from a 2-year-old plantation and 3,300 kg from a 3-year-old plantation.
Selling price	:	The average market price of dried roots is Rs 681.61/kg

Shankhpushpi



Scientific name	**:**	*Clitoria ternatea*
Common name	:	Butterfly pea flower, Blue pea vine (English), Aparajita (Hindi and Sanskrit), Billisaiuga, Satugadagida (Kannada)
Growth Habit	**:**	It is a perennial herbaceous vine / creeper; leaves are elliptic / obtuse; flowers are solitary, deep blue or white in colour with light yellow markings, flat pods with six to ten black seeds in each pod; roots contain rhizobium and which fix atmospheric nitrogen.
Economic Part	**:**	Whole plant including leaves, stem and roots are used
Medicinal Uses	**:**	Improves memory, enhance concentration, learning capabilities, used in treating mental fatigue, insomnia, insanity, anxiety, depression, asthma, bronchitis, constipation, ulcers and rejuvenation of the mind and nervous system.
Herbal Preparations	**:**	Tablets / capsules, powder / churna, juice / kashayam, syrup / arishtam, oil, tailam

Cultivation practices

Varieties	**:**	No named varieties
Climate	**:**	Tropical and sub-tropical climate; moderately shade-tolerant but can normally grow in full sunlight; temperature 25 - 30 °C
Soil	:	It can grow in variety of soil ranging from black cotton soil to loamy soil having good drainage system. However, red sandy loam soil having pH 5.5-7 is best for the crop.
Propagation	**:**	Seeds (400 g/ ha)
Spacing and planting	**:**	The seeds are sown with the onset of monsoon (I week of July) at a row spacing of 45 cm.
Manuring	**:**	10 tonnes FYM/ha as basal dose before sowing
Harvesting and Yield	**:**	First harvesting is done 3 month after planting by cutting at the ground level. Two subsequent cuttings are obtained at 90 days intervals. Average dry biomass yield of 10 tonnes can be obtained per hectare area. The herbage should be dried under shade to preserve the green colour and quality.
Selling price	**:**	The average market price of dried roots is Rs 681.61/kg

Shatavari



Scientific name	:	*Asparagus racemosus*
Common name	:	Asparagus (English), Satawar, Satavari (Hindi), Majjige – gedde, Aheruballi (Kannada), Satamuli, Satavari (Sanskrit),
Growth Habit	:	Tall perennial scandant and rumbling excessively branched spinous under shrub with tuberous roots
Economic Part	:	Roots
Medicinal Uses	:	Root is demulcent, diuretic, aphrodisiac, tonic, alterative, antiseptic, antidiarrhoeal, galactogogue and antispasmodic. Helps in cooling down the body from summer heat, cures hyper-acidity and peptic ulcer
Herbal Preparations	:	Shatavarighrtam, Shatavari juice, Shatavari churna, Shatavari extract, Shatavar capsule

Cultivation practices

Varieties	:	CIM - Shakti
Climate	:	Temperate to tropical and sub-tropical hilly regions where the elevations is between 800 to 1500 m above MSL are suitable
Soil	:	Prefers moist lateritic, red sandy loamy soils, with adequate drainage
Propagation	:	Root suckers, tuberous roots and seeds (1 kg/ha)
Spacing and planting	:	Seeds are usually soaked in warm water for a day before sowing and they usually start germinating after 40 days. Planted in the main field at spacing of 60 x 60 cm with onset of monsoon
Manuring	:	20 tonnes FYM/ha
Harvesting and Yield	:	The roots come to maturity in about 12-14 months after planting. A single plant may yield about 500 to 600 g of fresh root. On an average, 12,000 to 14,000 kg of fresh roots can be harvested from one-hectare area which on drying may yield about 1000 to 1200 kg of dried roots.
Selling price	:	The average market price of dried roots is Rs. 262/kg

Stevia



Scientific name	:	*Stevia rebaudiana*
Common name	:	Honey plant, Sweet leaf, Sugar leaf
Growth Habit	:	Slender perennial herb growing up to a height of 60 to 70 cm with sweet leaves
Economic Part	:	Leaves
Medicinal Uses	:	Used as sweetener in confectionery, bakery, beverage, medicine, in making chewing gum, mints and refreshers, used in herbal medicine and tonics for diabetic patients, in tooth pastes, mouth fresheners, soft drinks, pastry, pickles, tobacco products, candy, jam chewing gum due to its antibacterial and antifungal properties
Herbal Preparations	:	Dry leaves and dry leaf powder

Cultivation practices

Varieties	:	CIM-Madhu, MDS-13, MDS-14
Climate	:	Semi-humid subtropical plant and can grow in the temperature ranges between 11-41°C
Soil	:	Prefers sandy loam acidic to neutral soil with a pH range of 6.5-7.5
Propagation	:	Stem cuttings
Spacing and planting	:	Stem cuttings of 15 cm length are planted at spacing of 45 x 22 cm in the month of Feb-March.
Manuring	:	50 tonnes FYM/ha
Harvesting and Yield	:	The crop attains the stage of first harvest after three months of planting and subsequent harvests can be taken at 90 days interval. On an average, 3.0-3.5 tonnes of dry leaf yield can be obtained per harvest from one hectare. Totally a yield of 10 to 12 tonnes of dried leaves can be obtained in one year from one-hectare area.
Selling price	:	Not available

Sweet Flag



Scientific name	:	*Acorus calamus*
Common name	:	Sweet flag (English), Bach, Vach, Vadaja (Hindi), Bajegone (Kannada), Bhutanashini (Sanskrit)
Growth Habit	:	Monocot perennial semi-aquatic herb with narrow 0.75-1.5 m long leaves.
Economic Part	:	Leaves, stem and rhizome
Medicinal Uses	:	Digestive, cures constipation, gas, bloating, colic expectorant; to improve memory power and intellect, aids in clearing speech, cures, asthma, cough, bronchitis, diarrhoea, dysentery, infantile fever, insanity, hysteria and chronic rheumatic complaints.
Herbal Preparations	:	Vacha churna, Vacha capsule

Cultivation practices

Varieties	:	CIM-Balya
Climate	:	Prefers an area having a good and well distributed rainfall all throughout the year.
Soil	:	Prefers clayey loams and light alluvial soils with high moisture holding capacity
Propagation	:	Rhizome (400kg/ha)
Spacing and planting	:	Rhizomes are planted at spacing of 30x30 cm coinciding with March-April
Manuring	:	15 tonnes FYM/ha
Harvesting and Yield	:	Crop is ready for harvest one year after planting. The yield of rhizomes is about 10 tonnes per hectare.
Selling price	:	The average market price for rhizomes is Rs 121/kg

Velvet bean



Scientific name	:	*Mucuna pruriens*
Common name	:	Cowhedge (English), Kounch, Kewanch (Hindi), Nasugunni (Kannada), Kapikacchu (Sanskrit)
Growth Habit	:	Annual climber
Economic Part	:	Seeds
Medicinal Uses	:	Anti-parkinson, against hypertension, anti-diabetic, aphrodisiac, nervine tonic, useful in scorpion sting, leucorrhoea, spermatorrhoea and menstrual disorders
Herbal Preparations	:	Mucuna churna, Mucuna tablets

Cultivation practices

Varieties	:	Arka Dhanvantari, Arka Aswini, Arka Shubhra, Arka Shukla, Arka Daksha, Arka Charaka,CIM-Ajar
Climate	:	Grows under tropical and sub-tropical areas, also comes under drought conditions
Soil	:	Prefers well drained sandy to clayey loam type of soils
Propagation	:	Seeds (20-25 kg/ha)
Spacing and planting	:	Seeds are sown in June - July months at a spacing of 60 x 45 cm
Manuring	:	10 tonnes of FYM/ha
Harvesting and Yield	:	Fruits are harvested 5-6 months after sowing**.** Dry pods are harvested 3 to 4 times in the season. Gives average yield of 1.50- 1.75 t/ha (without staking); 3.00-3.75 t/ha (with staking); seed crop with staking: 5 t/ha
Selling price	:	The average market price of seeds is Rs. 100/kg

Tulsi



Scientific name	:	*Ocimum sanctum*
Common name	:	Holy basil (English), Tulsi (Hindi, Kannada, Sanskrit)
Growth Habit	:	Erect, herbaceous, much-branched, softly hairy, biennial or triennial, growing up to a height of 30-75 cm
Economic Part	:	Whole plant
Medicinal Uses	:	Possess anti-fertility, anticancer, antidiabetic, antifungal, antimicrobial, cardioprotective, analgesic, antispasmodic, adaptogenic properties and also used for the treatment of bronchitis, malaria, skin disease, common colds, headaches, stomach disorders, inflammation, arthritis, eye diseases, insect bites etc
Herbal Preparations	:	Tulsi tea, Tulsi powder, Tulsi kadha, Tulsi drop

Cultivation practices

Varieties	:	CIM-Ayu, CIM-Angana, Kanchan
Climate	:	Tropical and sub-tropical conditions with long day, high temperature
Soil	:	Thrives on rich loam to poor laterite, saline and alkaline to moderately acidic soils
Propagation	:	Seeds (200-300 g/ha)
Spacing and planting	:	Seeds are raised in nursery in the third week of February and seedlings are ready for transplanting in about 6 weeks' time at 4-5 leaf stage in the middle of April. Seedlings are spaced at 40 x 40 cm, 40 x 50 cm or 50 x 30 cm spacing.
Manuring	:	15 tonnes FYM/ha
Harvesting and Yield	:	Crop is harvested at full bloom stage. First harvest is obtained at 90-95 days of planting. Then it may be harvested every 65-75 days interval. About 5 tonnes of fresh herbage can be obtained twice or thrice a year per hectare.
Selling price	:	Average market price of whole herb is Rs 90/kg and seeds is Rs 132/kg

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